University Library Analytics – Data Warehouse Project
Group 1

1. Introduction

This report outlines the individual contributions of each team member, the challenges encountered during the implementation of the *University Library Analytics Data Warehouse*, and the key lessons learned throughout the project. The project aimed to design and implement a complete data warehouse solution integrating heterogeneous data sources, enabling analytics, OLAP operations, and decision support for university library management.

2. Individual Contributions

Team Member 1: Data Architect & Database Designer

Key Responsibilities Completed:

- Designed the overall **data warehouse architecture** using a **bottom-up (Kimball) approach**.
- Selected a **3-tier architecture** (Staging → Data Warehouse → Analytics).
- Designed the **dimensional model** using a **Star Schema**.
- Created all **dimension tables** (Time, Student, Department, Resource, Room).
- Designed and implemented the **fact table (library_usage_fact)** integrating data from all three sources.
- Defined **surrogate keys**, **primary keys**, and **foreign key relationships**.
- Wrote and tested **SQL DDL scripts** for:
  - Database creation
  - Staging tables
  - Dimension tables
  - Fact tables
  - Indexes for performance optimization
- Produced **ERD and schema diagrams**.
- Prepared a **data dictionary** documenting tables, columns, and data types.

- Justified the choice of **Star Schema** over Snowflake for query performance and simplicity.

Outcome:
A well-structured, scalable, and optimized data warehouse schema that supports efficient OLAP queries and reporting.

Team Member 2: ETL Specialist & Data Quality Manager

Key Responsibilities Completed:

- Designed and implemented the **ETL pipeline** using **Python and Pandas**.
- Extracted data from:
  - MySQL database (book transactions)
  - Excel files (digital usage)
  - CSV files (room bookings)
- Created a **staging area** for raw data validation.
- Implemented comprehensive **data cleansing rules**, including:
  - Standardizing department and faculty names (e.g., CS → Computer Science)
  - Converting all dates to **ISO 8601 (YYYY-MM-DD)** format
  - Removing duplicate and near-duplicate records
  - Standardizing room numbers and time slots
  - Handling missing values using flags, NULLs, and default records (e.g., “Unknown” student)
- Validated data types and ranges before loading into the warehouse.
- Implemented **error logging and validation checks**.
- Produced a **Data Quality Report** showing before/after improvement metrics.
- Documented all transformation rules and ETL logic.
- Defined **data refresh schedules** and incremental load strategy.

Outcome:
Clean, consistent, and reliable data loaded into the warehouse, significantly improving data quality and trustworthiness.

Team Member 3: Analytics, Reporting & Security Lead

Key Responsibilities Completed:

- Designed and implemented **OLAP operations**, enabling:
    - Drill-down (Year → Quarter → Month → Day)
    - Roll-up (Day → Week → Month)
    - Slicing and dicing across multiple dimensions
    - Pivot analysis for department and resource comparisons
- Developed **10+ SQL analytical queries** to answer all business questions.
- Built **three interactive dashboards**:
    - Executive Dashboard (KPIs and trends)
    - Department Dashboard (usage by faculty)
    - Operational Dashboard (daily activity and peak times)
- Implemented **Role-Based Access Control (RBAC)**:
    - Admin role (full access)
    - Analyst role (read-only)
    - Department Head role (department-specific access)
- Applied **data masking** for sensitive fields such as StudentID.
- Implemented **security best practices**, including access auditing.
- Designed a **backup and recovery plan** (full and incremental backups).
- Prepared a **user guide** for non-technical stakeholders.

Outcome:
A secure, user-friendly analytics environment that supports data-driven decision-making.

3. Challenges Faced

1. **Data Inconsistencies**
    - Different naming conventions for departments, rooms, and resource types required extensive standardization.
2. **Multiple Date Formats**

- Handling inconsistent date formats across Excel, CSV, and database sources required careful parsing and validation.

3. **Missing and Incomplete Data**
    - NULL values for ReturnDate, StudentID, and Duration_Minutes needed clear business rules to avoid misleading analytics.
4. **Duplicate Records**
    - System-generated duplicates required both exact and logical duplicate detection rules.
5. **Integration of Heterogeneous Data Sources**
    - Combining relational databases, spreadsheets, and CSV files into a unified schema was technically challenging.
6. **Security Implementation**
    - Designing RBAC while still allowing meaningful analytics required careful role definition.

4. Lessons Learned

- **Data quality is critical**: Poor-quality input data leads to unreliable analytics, regardless of system design.
- **Dimensional modeling simplifies analytics**: Star Schema significantly improved query readability and performance.
- **ETL planning saves time**: Clearly defined transformation rules reduced errors during implementation.
- **Collaboration is essential**: Clear role division improved efficiency and reduced overlap.
- **Documentation matters**: Well-documented processes made troubleshooting and presentation easier.
- **Security cannot be an afterthought**: Integrating access control early avoided redesign later.

5. Conclusion

This project provided hands-on experience in designing and implementing a real-world data warehouse solution. By integrating multiple data sources, applying robust ETL processes, enabling OLAP analysis, and ensuring security and data quality, the team successfully delivered a scalable and reliable **University Library Analytics Data**